ATOMQUEST 2025- ABSTRACT SUBMISSION

PROJECT TITLE:- CIRCULAR QUDRANT BASED ROBOTICS CLEANING BOT

1. Team Information

• **Team Name:** Super Kings

• **Team Members:** Harsh Raj, Rakesh Hembram, Gajendra

• **Institute:** Indian Institute of Technology, Patna

2. Institute Resources

We will utilize the following facilities at our institute to develop the prototype:

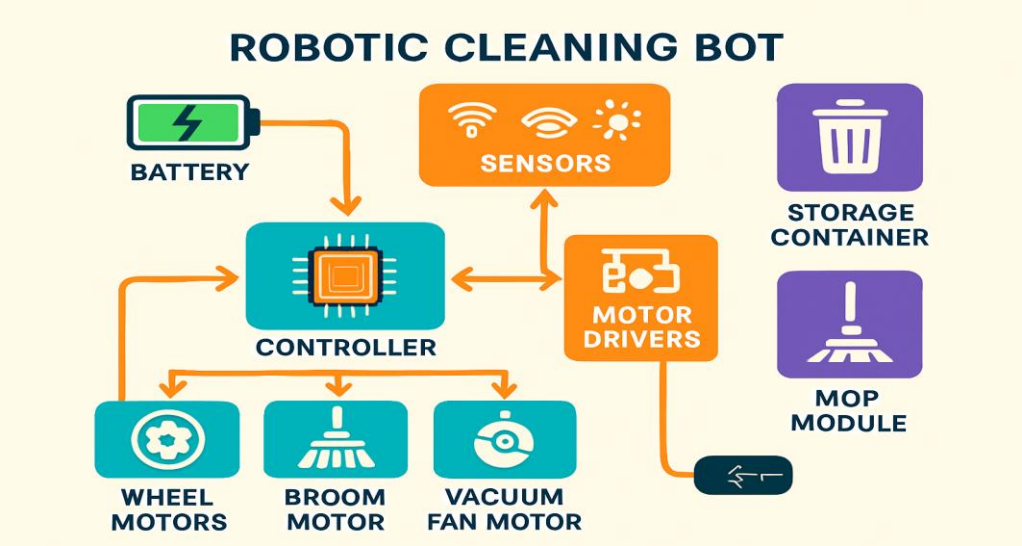
- Robotics Lab
- Mechanical Workshop
- Electronics & Communication Lab
- CAD/Simulation Lab
- Micro fabrication lab
- Trilogy lab

3. Proposed Solution Overview

Our team proposes a circular robotic cleaning bot with a unique quadrant-based design. Each quarter of the circular chassis is equipped with a specific cleaning mechanism: a rotating broom brush for dry waste, a vacuum suction unit for fine dust, and a mop module for wet cleaning. The fourth quarter houses the dust storage container and electronic components. This design enables versatile cleaning in a single pass, ensuring higher efficiency and adaptability in office environments.

4. System Level Block Diagram

Below is a rough schematic of the system design:



5. Rough Specifications

Shape	Circular (≤ 1.5 ft diameter, ≤ 1 ft height)
Battery	12V Li-ion, ~4000 mAh (30–40 min runtime)
Motors	2 DC drive motors, 1 vacuum motor, 1 mop motor, 1 brush motor
Controller	Arduino Uno / Raspberry Pi
Sensors	Ultrasonic, IR, Dust sensor
Wheels	2 powered wheels + 1 caster
Dust Storage	~300–400 ml removable container
Chassis Material	ABS Plastic / Lightweight Aluminum
Weight	Approx. 4–5 kg
Cleaning Mechanism	Hybrid – broom, suction, mop
Container Handling	Latch-based removable storage

6. Rough Component List

- DC Motors
- Li-ion Battery Pack
- Ultrasonic/IR Sensors
- Arduino / Raspberry Pi Controller
- Vacuum Motor / Fan Unit
- Mop Roller
- Plastic/Metal Chassis
- Wheels + Caster Wheel
- Dustbin Container

7. Unique Features

- Hybrid cleaning in one pass (broom + vacuum + mop)
- Circular shape for smooth maneuverability
- Modular quadrant design for easy upgrades
- Energy-efficient operation



